Wenkai Qin

CONTACT INFORMATION

Department of Aerospace Engineering and Engineering Mechanics
The University of Texas at Austin
2617 Wichita Street, Desk 3.04
Austin, TX 78712
Email: wqin@utexas.edu
Mobile: (203) 914-2341

EDUCATION

University of Texas at Austin, Austin, Texas USA
Ph.D., Aerospace Engineering, May 2027 (expected)
Advisor: Todd E. Humphreys
Affiliation: Radionavigation Laboratory

University of Texas at Austin, Austin, Texas USA
M.S., Aerospace Engineering, May 2024 (expected)

Harvey Mudd College, Claremont, California USA
B.S., Engineering, May 2018
Concentration: Japanese
Graduated with distinction

ACADEMIC POSITIONS

Research Assistant, University of Texas at Austin — September 2022 - present

OTHER PROFESSIONAL EXPERIENCE

Raytheon Missiles and Defense — September 2019 - April 2022
Guidance Navigation and Control Engineer II — Tucson, AZ
- Developed optimal guidance, navigation, and control laws for use on defense interceptors.
- Captured and interpreted physical flight test data to improve 6DOF simulation fidelity.
- Designed exo-atmospheric pointing algorithm, cutting per-event cold gas usage by over 30%.
- Developed autonomous horizon-based navigation algorithm (US Patent No. US11580690B1).

Laserfiche — February 2019 - September 2019
Data Engineer — Long Beach, CA
- Led project to spin up in-house business intelligence data warehouse.
- Developed visualizations, interactive business intelligence reports for company executive leadership.

C2 Education — July 2018 - February 2019
Private Tutor — Greenwich, CT
- Tutored children ranging from age 10 to 18 in subjects including calculus, physics, and chemistry.
- Provided suggestions to parents based on regular progress reports specific to the child's needs.

MEMBERSHIPS IN PROFESSIONAL AND HONORARY SOCIETIES

Member, Institute of Navigation (ION)

PUBLICATIONS

Refereed Conference Proceedings

C1 W. Qin, Z. M. Komodromos, and T. E. Humphreys, "An agile, portable antenna system for LEO megaconstellation-based PNT," in *Proceedings of the ION GNSS+ Meeting*, 2023 [pdf]

C2 Z. M. Komodromos, W. Qin, and T. E. Humphreys, "Signal simulator for Starlink Ku-Band downlink," in *Proceedings of the ION GNSS+ Meeting*, 2023 [pdf]

Patents

P1 J. S. K. James N. Head, Wenkai Qin and J. D. Verret, "Horizon-based navigation," February 2023. US Patent 11,580,690 B1 [link]

TEACHING ACTIVITIES

1. Spring 2015: Graded and tutored for CS5 (Introduction to Computer Science) at Harvey Mudd College
2. Fall 2015: Graded and tutored for Physics 24 (Mechanics and Wave Motion) at Harvey Mudd College
3. Fall, Spring 2018: Graded and tutored for E101-102 (Advanced Systems Engineering I-II) at Harvey Mudd College